PROJECT TITLE: GRAMSETU

The Unified Rural Civic-Tech Gateway & Agronomic Intelligence Platform

File Naming: GramSetu – Team Eclipse **Team Name:** Team Eclipse (Lead: Eshwar H S) **Category:** IT Domain

1. Problem Statement

Over 65% of the national population resides in rural regions, yet rural citizens encounter pervasive structural exclusion from modern e-governance systems and agricultural technology. In major farming states like Karnataka, over 6 crore rural citizens and smallholder farmers face acute operational bottlenecks:

1.1 Rural Administrative & Linguistic Divide

Existing governmental digital portals mandate high digital literacy, intricate multi-stage keyboard typing, and English fluency. Rural farmers and elderly villagers are systematically disenfranchised by complex authentication steps, bureaucratic jargon, and the complete absence of voice-guided vernacular navigation.

1.2 Agronomic Information Asymmetry & Crop Vulnerability

Smallholder farmers suffer severe annual crop yield losses (20%–35%) due to delayed pathogen diagnosis and uncalibrated chemical pesticide usage. Extension officers cannot feasibly provide timely on-field inspections across remote villages, leaving farmers vulnerable to vendor exploitation.

1.3 Escalation Vacuum In Village Grievance Redressal

Critical village infrastructure failures—water pipeline leaks, transformer burnouts, and canal blockages—languish indefinitely due to paper filings, absence of location-aware tracking, and lack of enforceable Service Level Agreement (SLA) escalation hierarchies.

2. Measurable Project Objectives

GramSetu establishes benchmarked technological goals to bridge the rural governance divide:

- **2.1 Sub-Second Multilingual Voice Interaction:** Deliver speech-to-speech interaction in regional languages and English with audio synthesis latency under 800ms and conversational intent accuracy exceeding 92%.
- **2.2 High-Precision Real-Time Crop Diagnostics:** Achieve computer vision pathology diagnosis for 15+ staple crops with > 90% accuracy and inference under 2.5s via Google Gemini Multimodal Vision with offline classifier fallback.
- **2.3 Automated 4-Tier SLA Grievance Resolution:** Programmatically enforce strict escalation timelines (Gram Panchayat PDO: 7d → Taluk EO: 14d → ZP CEO: 21d → State Comm.) reducing grievance resolution cycle time by 60%.
- **2.4 Real-Time APMC Mandi Market Transparency:** Aggregate live daily APMC price quotations across 31 districts with intraday volatility indicators to protect farmers from middleman exploitation.

3. Methodology

3.1 Working Principle & Architectural Logic

GramSetu is engineered as a reactive, voice-first single-page progressive web application. User vocal commands are captured via browser Web Audio & Speech APIs, streamed into an intelligent intent router, and dispatched to either client controllers or Google Gemini generative models.

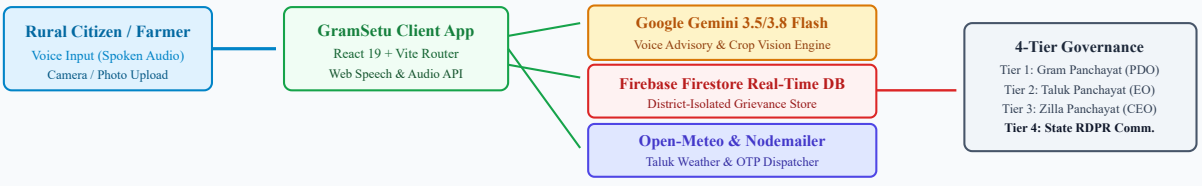


Figure 1: GramSetu End-to-End System Architecture, AI Pipelines & Hierarchical Governance Flowchart

3.2 AI Model Pipelines & Offline Fallback Engineering

For crop disease scanning, camera captures are compressed on the client via HTML5 Canvas into base64 frames and evaluated against Google Gemini 3.5 Flash Vision. In disconnected environments or during API rate limits, a client-side Vision Feature Classifier automatically steps in, performing local pixel color distribution analysis (chlorosis, necrosis, mildew) to deliver instant agricultural triage.

4. Scope And Limitations

4.1 Operational Scope

GramSetu encompasses 31 administrative districts, 240+ taluks, and 6,000+ Gram Panchayats. The crop database is fine-tuned for staple commercial crops (Paddy, Ragi, Arecanut, Maize, Cotton, Tomato, Sugarcane) across diverse agro-climatic zones.

4.2 Technical Constraints & Mitigations

- **Low Bandwidth Environments:** Handled via client-side Canvas JPEG compression (< 200KB payloads) and localized offline JSON datasets.
- **Acoustic Ambient Field Noise:** Outdoor wind and tractor noise during speech recognition is counteracted by acoustic transcript normalization and phonetic transliteration algorithms.
- **Hardware Neutrality:** Operates entirely on standard mobile and desktop browsers with zero requirement for specialized hardware.

5. Project Features

GramSetu consolidates four core functional pillars into a unified, responsive civic portal:

5.1 Crop Doctor — AI Pathology Scanner

Instant visual crop diagnosis via Gemini Vision. Generates actionable reports: severity ratings, chemical dosages (Hexaconazole, Mancozeb), organic recipes, and direct insurance compensation links.

5.2 GramSetu AI Voice Assistant

Multilingual speech-to-speech assistant supporting regional voice commands. Provides hands-free portal navigation and answers MSP, weather, and fertilizer questions with live waveforms.

5.3 4-Tier Automated Escalation Ledger

Location-scoped grievance submission (District → Taluk → GP). Features photo evidence, automated SLA clock, tamper-proof tracking, and official administrative consoles.

5.4 Live APMC Mandi Intelligence

Live district-wise APMC pricing for Arecanut, Copra, Ragi, Paddy, and Turmeric. Simulates intraday tick movements with visual bullish/bearish indicators and 5-day trends.

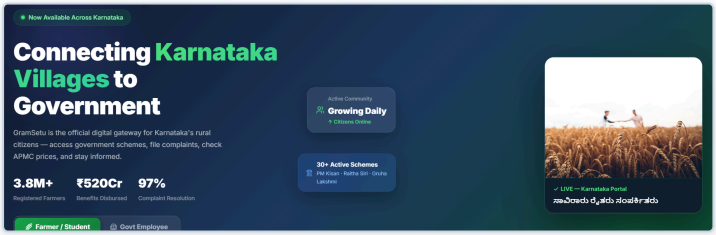


Figure 2: GramSetu Production Working Model — Live Citizen Portal

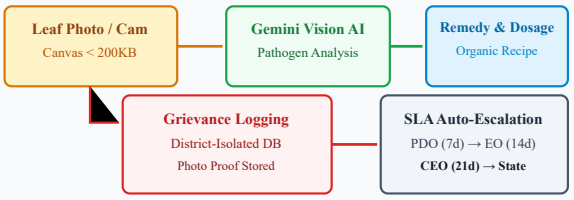


Figure 3: Dual AI Diagnostic & Escalation Flowchart

6. Tech Stack

Layer / Component	Technologies & Frameworks Utilized	Operational Purpose
Frontend Architecture	React 19, Vite 8, React Router v7, Vanilla CSS, Lucide React	High-performance, ultra-responsive SPAs with zero bundle bloat
Artificial Intelligence	Google Gemini 3.5/3.8 Flash, Gemini Vision, Web Speech API	Speech-to-Speech Q&A, Multimodal Plant Pathology & Advisory
Backend & Database	Firebase Firestore (Realtime NoSQL), Firebase Auth	District-isolated document storage & live data synchronization
APIs & Cloud Services	Open-Meteo Weather API, Nodemailer Gmail SMTP, Vercel Serverless	Real-time taluk forecasts, secure OTP emails, global edge deployment
Reporting & Visuals	Leaflet Maps, jsPDF, Puppeteer, Web Audio API	Interactive GIS heatmaps, official PDF exports, dynamic audio waves

7. Innovativeness

GramSetu introduces key architectural paradigm shifts that distinguish it from legacy e-governance systems:

- **7.1 Multimodal Voice-First Rural Interaction:** Eliminates keyboard literacy requirements through end-to-end hands-free voice navigation and natural conversational speech synthesis in vernacular accents.
- **7.2 Hybrid Edge-Cloud AI Resilience:** Combines Google Gemini Vision with an in-browser heuristic pixel classifier, guaranteeing that farmers in remote fields receive immediate crop triage even without network connectivity.
- **7.3 Automated Algorithmic Accountability:** Replaces discretionary bureaucratic handling with an automated 4-tier countdown escalation system, stripping local officials of the ability to bury citizen complaints without higher-tier oversight.
- **7.4 Hyper-Local Geographic Scoping:** Automatically filters data to "My Taluk" and "My Gram Panchayat", eliminating statewide clutter for rural citizens while isolating administrative domains for officials.

8. Market Potential

The primary addressable base spans 6.1+ crore rural citizens, 78+ lakh agricultural households, and 6,000+ Gram Panchayats. Commercial and institutional viability includes:

- **Government Civic-Tech Procurement:** Direct institutional adoption by the Department of Rural Development & Panchayat Raj and Department of Agriculture for digitized grievance redressal and agricultural extension.
- **Farmer Producer Organizations (FPOs):** Subscription deployment for FPOs to broadcast APMC prices, manage collective grievances, and diagnose member crops.
- **Socio-Economic Impact Across Farming Communities:** Rapid crop triage saves 15% to 25% harvest value per acre while algorithmic SLA escalation empowers rural citizens with institutional accountability.

9. Scalability

9.1 Architectural Cloud Elasticity & Sharded Persistence

Built on horizontally scalable serverless primitives (Vercel Edge + Firebase distributed NoSQL). The schema utilizes district-sharded collections (e.g. `district_{districtName}_complaints`), ensuring database index contention remains O(1) regardless of concurrent statewide traffic surges.

9.2 Pan-India Multilingual Localization & Progressive Web Capabilities

Linguistic string tables are decoupled into atomic JSON stores. Adapting to other agricultural states requires solely adding regional lexicons (Marathi, Telugu, Hindi) and local APMC mandi tables, running smoothly on budget smartphones without native app installs.

10. Other Project Details

The complete, functional production system is deployed live on cloud infrastructure and fully open-source for academic and jury evaluation:

10.1 Project Hosting Link (Live Working Application):
• Live Web URL: **https://gram-setu-one.vercel.app**

10.2 Project Hosted Source Code Link (GitHub Repository):
• GitHub Repository: **https://github.com/eshwarhs170-a11y/GramSetu**

10.3 Reference Links & APIs:
• Google Gemini Developer API & Documentation: **https://ai.google.dev**
• Google DeepMind Generative Language Platform: **https://deepmind.google/technologies/gemini/**
• Firebase Realtime Cloud Firestore: **https://firebase.google.com**
• Open-Meteo Global Numerical Weather Forecast API: **https://open-meteo.com**
• State Rural Development & Panchayat Raj Administrative Portal: **https://rdpr.karnataka.gov.in**
• PM Kisan Samman Nidhi & PMFBY National Portals: **https://pmkisan.gov.in**